

	17	Bolzano's Intermediate Value Theorem,		
	18	Uniform Continuity,		
	19	Monotone and Inverse Functions		
	Sections from References:			
IV			10	15
	20	Derivative		
	21	Chain Rule		
	22	The Mean Value Theorem		
	23	Riemann Integral, Riemann Integrable Functions,		
	24	Fundamental Theorem of Calculus		
	Sections from References:			
V	Open Ended Module:		12	
	1	Sets and Functions, Finite and Infinite Sets Algebraic Properties of R, Rational and Irrational Numbers,		
	Sections from References:			
Books and References:				
1. Bartle R. G. and Sherbert D. R. (2000). Introduction to Real Analysis, 3 rd edition, John Wiley & Sons				
2. Rudin, W. (1976) Principles of Mathematical Analysis, McGraw-Hill, New York.				
3. Royden, H. L. and Fitzpatrick, P. M. (2010). Real Analysis. Prentice Hall.				

Programme	B. Sc. Statistics				
Course Code	STA3CJ202(P)/STA3MN200 (P)				
Course Title	Probability and Random Variables				
Type of Course	Major				
Semester	III				
Academic Level	200-299				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	HSE level Mathematics/Statistics courses				
Course Summary Objective	Make the students recall set theory, define, classify, illustrate probability theory. Discuss use of math. Expectation in variable properties				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Define and comprehend the fundamental concepts of probability and apply basic	U	C	Instructor-created exams / Quiz

	probability rules.			
CO2	Define random variables, compute their probabilities, and consequently develop probability and cumulative probability distributions.	Ap	P	Practical Assignment / Observation of Practical Skills
CO3	Recognize and interpret moments of a distribution through mathematical expectation	Ap	P	Seminar Presentation / Group Tutorial Work
CO4	Develop proficiency in handling probability problems using statistical software and analyzing probability distributions.	U	C	Instructor-created exams / Home Assignments
CO5	Communicate the solutions to probability problems effectively and enhance the ability to present information clearly and concisely	Ap	P	One Minute Reflection Writing assignments
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I	Basics of Set Theory		12	18
	1	Permutations and Combinations		
	2	Random experiment, Sample space, Events,		
	3	Classical definition of probability		
	4	Statistical regularity		
	5	Statistical Definition of Probability		
	6	Field, Sigma field		
	7	Axiomatic definition of probability and simple properties		
	8	Addition theorem (two and three events)		
	Sections from References:			
II	Conditional probability		10	15
	9	Definition of Conditional probability		
	10	Multiplication theorem		
	11	Independence of events- Pair wise and Mutual		
	12	Bayes theorem and its applications.		

	Sections from References:			
III	Random variables		13	20
	13	Discrete and Continuous Random variables		
	14	Probability mass function (pmf) properties and examples		
	15	Probability density function (pdf)-properties and examples		
	16	Cumulative distribution function		
	17	Properties of Distribution Function		
	18	Plotting step function/Ladder function		
	19	Change (transformation) of variables		
	20	Derivative method		
	21	Distribution function method		
	Sections from References:			
IV	Mathematical Expectation		10	17
	22	Expected values of Random Variables		
	23	Raw and Central Moments (definition and relationships)		
	24	Moment generation function (MGF)		
	25	Properties of MGF		
	26	Characteristic function (definition and use only),		
	27	Moment measures of Skewness and kurtosis.		
	Sections from References:			
V	Open Ended Module: Practical Applications Probability and Distributions		30	
	1	Handling problems related to probability Verification of function as pmf/pdf, Evaluation of moments, MGF and characteristic function (R / Mathematica-wolframcloud /sage...) Case Study Observing a random phenomenon, construction of empirical probability distribution.		
	Sections from References:			